

Qns	Answer	Marks
6(ai)	Lamination increases resistance, thus <u>reduces induced (eddy) current in the iron core</u> , which would reduce heat loss.	M1
6(aii)	Alternating current in primary coil (I_{pri}) give rise to changing magnetic flux (ϕ) in the primary coil and the iron core. Thus primary coil current is in phase with the magnetic flux. This changing magnetic flux (ϕ) links the secondary coil through the iron core. E.m.f. induced in secondary coil (E_{sec}) is proportional to <u>rate</u> of change of flux (ϕ). Secondary coil e.m.f. (E_{sec}) is not in phase with the flux (ϕ). (Hence e.m.f induced in secondary coil not in phase with alternating current in primary coil.	B1 B1 B1

6(b)	$\left(\frac{V_{sec}}{V_{pri}}\right)_{rms} = \frac{N_{sec}}{N_{pri}}$ $(V_{sec})_{rms} = \frac{8100}{300} \times 9.0 = 243 \text{ V}$ $\text{Peak voltage across load} = (V_{sec})_{rms} \times \sqrt{2} = 243 \times \sqrt{2}$ $= 3.4 \times 10^2 \text{ V}$	C1 A1

Qns	Answer	Marks
7(a)(i)	The force on the electron is (always) perpendicular to the velocity/	B1